

4(a)	any four from: 1 geographical, isolation / barrier (caused by the, river / canyon walls) ; 2 no gene flow between (isolated ancestral) populations ; 3 different, selection pressures / environment ; 4 different / random / independent, mutations (in each population) ; 5 different changes in allele frequencies / different gene pools / different alleles selected for ; 6 populations have different, morphological / physiological / behavioral, features ; 7 leads to, reproductive isolation / inability to interbreed (to produce fertile offspring) ; 8 allopatric speciation ;	4
4(b)	any three from: <i>DNA sequencing ora for fossils</i> 1 provides more information (about the whole organism) ; 2 more precise / more accurate ; 3 only a small sample of DNA is needed ; 4 can give estimate of when species diverged/how closely related two organisms are / molecular clock can estimate time of divergence ; 5 quantitative ; 6 AVP ; e.g. <i>idea of.</i> avoids convergent evolution problems	3

Question	Answer	Marks
4(c)	<i>any one from:</i> not enough fossils ; small sample sizes ; DNA degrades over time ; this depends on an estimated mutation rate ;	1

Question	Answer	Marks
----------	--------	-------